SQL Benchmark Dataset Preparation

Objective:

The goal of this task is to prepare a clean benchmark dataset that will later be used to evaluate the Text-to-SQL agent system.

I will work with a predefined collection of natural language questions and manually create correct SQL queries for them. These SQL queries will act as the ground truth (reference answers) for evaluating whether the Text-to-SQL agent is generating correct outputs.

In addition, I will think critically about:
"How do we evaluate whether a Text-to-SQL agent is actually performing well?"
This task forms the foundation for all future assignments in this module.

(This document contains manually written correct SQL as well as outputs with explanations.)

Dataset

Use the following Google Sheet as the benchmark question dataset:

[SQL_Benchmark_Dataset]

Tables in text2sql_assignment:

```
postgres=# \c text2sql_assignment
You are now connected to database "text2sql_assignment" as user "postgres".
text2sql_assignment=# \dt
            List of relations
 Schema |     Name     | Type  |  Owner
--------+--------------+-------+----------
 public | customers    | table | postgres
 public | employees    | table | postgres
 public | offices      | table | postgres
 public | orderdetails | table | postgres
 public | orders       | table | postgres
 public | payments     | table | postgres
 public | productlines | table | postgres
 public | products     | table | postgres
(8 rows)

text2sql_assignment=#
```

SQL QUERY QUESTIONS

1. List all products

SQL Query:

```sql
SELECT * FROM products;
```

Query Result:

	productCode	productName	productLine	productScale	productVendor	productDescription	quantityInStock	buyPrice	MSRP
1	S10_1678	1969 Harley Davidson Ultimate Chopper	Motorcycles	1:10	Min Lin Diecast	This replica features working kickstand, front suspension, gear-shift lever,	7,933	48.81	95.7
2	S10_1949	1952 Alpine Renault 1300	Classic Cars	1:10	Classic Metal Creations	Turnable front wheels; steering function; detailed interior; detailed engine	7,305	98.58	214.3
3	S10_2016	1996 Moto Guzzi 1100i	Motorcycles	1:10	Highway 66 Mini Classics	Official Moto Guzzi logos and insignias, saddle bags located on side of mo	6,625	68.99	118.94
4	S10_4698	2003 Harley-Davidson Eagle Drag Bike	Motorcycles	1:10	Red Start Diecast	Model features, official Harley Davidson logos and insignias, detachable r	5,582	91.02	193.66
5	S10_4757	1972 Alfa Romeo GTA	Classic Cars	1:10	Motor City Art Classics	Features include: Turnable front wheels; steering function; detailed interic	3,252	85.68	136
6	S10_4962	1962 LanciaA Delta 16V	Classic Cars	1:10	Second Gear Diecast	Features include: Turnable front wheels; steering function; detailed interic	6,791	103.42	147.74
7	S12_1099	1968 Ford Mustang	Classic Cars	1:12	Autoart Studio Design	Hood, doors and trunk all open to reveal highly detailed interior features.	68	95.34	194.57
8	S12_1108	2001 Ferrari Enzo	Classic Cars	1:12	Second Gear Diecast	Turnable front wheels; steering function; detailed interior; detailed engine	3,619	95.59	207.8
9	S12_1666	1958 Setra Bus	Trucks and Buses	1:12	Welly Diecast Productions	Model features 30 windows, skylights & glare resistant glass, working ste	1,579	77.9	136.67
10	S12_2823	2002 Suzuki XREO	Motorcycles	1:12	Unimax Art Galleries	Official logos and insignias, saddle bags located on side of motorcycle, de	9,997	66.27	150.62
11	S12_3148	1969 Corvair Monza	Classic Cars	1:18	Welly Diecast Productions	1:18 scale die-cast about 10 long doors open, hood opens, trunk opens ar	6,906	89.14	151.08
12	S12_3380	1968 Dodge Charger	Classic Cars	1:12	Welly Diecast Productions	1:12 scale model of a 1968 Dodge Charger. Hood, doors and trunk all ope	9,123	75.16	117.44
13	S12_3891	1969 Ford Falcon	Classic Cars	1:12	Second Gear Diecast	Turnable front wheels; steering function; detailed interior; detailed engine	1,049	83.05	173.02
14	S12_3990	1970 Plymouth Hemi Cuda	Classic Cars	1:12	Studio M Art Models	Very detailed 1970 Plymouth Cuda model in 1:12 scale. The Cuda is gener	5,663	31.92	79.8
15	S12_4473	1957 Chevy Pickup	Trucks and Buses	1:12	Exoto Designs	1:12 scale die-cast about 20 long Hood opens, Rubber wheels	6,125	55.7	118.5
16	S12_4675	1969 Dodge Charger	Classic Cars	1:12	Welly Diecast Productions	Detailed model of the 1969 Dodge Charger. This model includes finely det	7,323	58.73	115.16

2. Get all customers

SQL Query:

```sql
SELECT * FROM customers;
```

Query Result:

	cus	customerName	contactLastName	contactFirstName	phone	addressLine1	addressLine2	city	state	postalCode	country	salesR
1	103	Atelier graphique	Schmitt	Carine	40.32.2555	54, rue Royale	[NULL]	Nantes	[NULL]	44000	France	1,370
2	112	Signal Gift Stores	King	Jean	7025551838	8489 Strong St.	[NULL]	Las Vegas	NV	83030	USA	1,166
3	114	Australian Collectors, Co.	Ferguson	Peter	03 9520 4555	636 St Kilda Road	Level 3	Melbourne	Victoria	3004	Australia	1,611
4	119	La Rochelle Gifts	Labrune	Janine	40.67.8555	67, rue des Cinquante Otages	[NULL]	Nantes	[NULL]	44000	France	1,370
5	121	Baane Mini Imports	Bergulfsen	Jonas	07-98 9555	Erling Skakkes gate 78	[NULL]	Stavern	[NULL]	4110	Norway	1,504
6	124	Mini Gifts Distributors Ltd.	Nelson	Susan	4155551450	5677 Strong St.	[NULL]	San Rafael	CA	97562	USA	1,165
7	125	Havel & Zbyszek Co	Piestrzeniewicz	Zbyszek	(26) 642-7555	ul. Filtrowa 68	[NULL]	Warszawa	[NULL]	01-012	Poland	[NULL]
8	128	Blauer See Auto, Co.	Keitel	Roland	+49 69 66 90 2555	Lyonerstr. 34	[NULL]	Frankfurt	[NULL]	60528	Germany	1,504
9	129	Mini Wheels Co.	Murphy	Julie	6505555787	5557 North Pendale Street	[NULL]	San Francisco	CA	94217	USA	1,165
10	131	Land of Toys Inc.	Lee	Kwai	2125557818	897 Long Airport Avenue	[NULL]	NYC	NY	10022	USA	1,323
11	141	Euro+ Shopping Channel	Freyre	Diego	(91) 555 94 44	C/ Moralzarzal, 86	[NULL]	Madrid	[NULL]	28034	Spain	1,370
12	144	Volvo Model Replicas, Co	Berglund	Christina	0921-12 3555	Berguvsvägen 8	[NULL]	Luleå	[NULL]	S-958 22	Sweden	1,504
13	145	Danish Wholesale Imports	Petersen	Jytte	31 12 3555	Vinbæltet 34	[NULL]	Kobenhavn	[NULL]	1734	Denmark	1,401
14	146	Saveley & Henriot, Co.	Saveley	Mary	78.32.5555	2, rue du Commerce	[NULL]	Lyon	[NULL]	69004	France	1,337
15	148	Dragon Souveniers, Ltd.	Natividad	Eric	+65 221 7555	Bronz Sok.	Bronz Apt. 3/6 Tesvikiye	Singapore	[NULL]	079903	Singapore	1,621
16	151	Muscle Machine Inc	Young	Jeff	2125557413	4092 Furth Circle	Suite 400	NYC	NY	10022	USA	1,286
17	157	Diecast Classics Inc.	Leong	Kelvin	2155551555	7586 Pompton St.	[NULL]	Allentown	PA	70267	USA	1,216
18	161	Technics Stores Inc.	Hashimoto	Juri	6505556809	9408 Furth Circle	[NULL]	Burlingame	CA	94217	USA	1,165
19	166	Handji Gifts& Co	Victorino	Wendy	+65 224 1555	106 Linden Road Sandown	2nd Floor	Singapore	[NULL]	069045	Singapore	1,612

3. Show all orders

SQL Query:

```sql
SELECT * FROM orders;
```

Query Result:

	orderNumber	orderDate	requiredDate	shippedDate	status	comments	customerNumber
1	10,100	2003-01-06	2003-01-13	2003-01-10	Shipped	[NULL]	363
2	10,101	2003-01-09	2003-01-18	2003-01-11	Shipped	Check on availability.	128
3	10,102	2003-01-10	2003-01-18	2003-01-14	Shipped	[NULL]	181
4	10,103	2003-01-29	2003-02-07	2003-02-02	Shipped	[NULL]	121
5	10,104	2003-01-31	2003-02-09	2003-02-01	Shipped	[NULL]	141
6	10,105	2003-02-11	2003-02-21	2003-02-12	Shipped	[NULL]	145
7	10,106	2003-02-17	2003-02-24	2003-02-21	Shipped	[NULL]	278
8	10,107	2003-02-24	2003-03-03	2003-02-26	Shipped	Difficult to negotiate with customer. We nee	131
9	10,108	2003-03-03	2003-03-12	2003-03-08	Shipped	[NULL]	385
10	10,109	2003-03-10	2003-03-19	2003-03-11	Shipped	Customer requested that FedEx Ground is u	486
11	10,110	2003-03-18	2003-03-24	2003-03-20	Shipped	[NULL]	187
12	10,111	2003-03-25	2003-03-31	2003-03-30	Shipped	[NULL]	129
13	10,112	2003-03-24	2003-04-03	2003-03-29	Shipped	Customer requested that ad materials (suc	144
14	10,113	2003-03-26	2003-04-02	2003-03-27	Shipped	[NULL]	124
15	10,114	2003-04-01	2003-04-07	2003-04-02	Shipped	[NULL]	172
16	10,115	2003-04-04	2003-04-12	2003-04-07	Shipped	[NULL]	424
17	10,116	2003-04-11	2003-04-19	2003-04-13	Shipped	[NULL]	381
18	10,117	2003-04-16	2003-04-24	2003-04-17	Shipped	[NULL]	148
19	10,118	2003-04-21	2003-04-29	2003-04-26	Shipped	Customer has worked with some of our ven	216

4. List all employees

SQL Query:

```sql
SELECT * FROM employees;
```

Query Result:

	employeeNumber	lastName	firstName	extension	email	officeCode	reportsTo	jobTitle
1	1,002	Murphy	Diane	x5800	dmurphy@classicmodelcars.com	1	[NULL]	President
2	1,056	Patterson	Mary	x4611	mpatterso@classicmodelcars.com	1	1,002	VP Sales
3	1,076	Firrelli	Jeff	x9273	jfirrelli@classicmodelcars.com	1	1,002	VP Marketing
4	1,088	Patterson	William	x4871	wpatterson@classicmodelcars.com	6	1,056	Sales Manager (APAC)
5	1,102	Bondur	Gerard	x5408	gbondur@classicmodelcars.com	4	1,056	Sale Manager (EMEA)
6	1,143	Bow	Anthony	x5428	abow@classicmodelcars.com	1	1,056	Sales Manager (NA)
7	1,165	Jennings	Leslie	x3291	ljennings@classicmodelcars.com	1	1,143	Sales Rep
8	1,166	Thompson	Leslie	x4065	lthompson@classicmodelcars.com	1	1,143	Sales Rep
9	1,188	Firrelli	Julie	x2173	jfirrelli@classicmodelcars.com	2	1,143	Sales Rep
10	1,216	Patterson	Steve	x4334	spatterson@classicmodelcars.com	2	1,143	Sales Rep
11	1,286	Tseng	Foon Yue	x2248	ftseng@classicmodelcars.com	3	1,143	Sales Rep
12	1,323	Vanauf	George	x4102	gvanauf@classicmodelcars.com	3	1,143	Sales Rep
13	1,337	Bondur	Loui	x6493	lbondur@classicmodelcars.com	4	1,102	Sales Rep
14	1,370	Hernandez	Gerard	x2028	ghernande@classicmodelcars.com	4	1,102	Sales Rep
15	1,401	Castillo	Pamela	x2759	pcastillo@classicmodelcars.com	4	1,102	Sales Rep
16	1,501	Bott	Larry	x2311	lbott@classicmodelcars.com	7	1,102	Sales Rep
17	1,504	Jones	Barry	x102	bjones@classicmodelcars.com	7	1,102	Sales Rep
18	1,611	Fixter	Andy	x101	afixter@classicmodelcars.com	6	1,088	Sales Rep
19	1,612	Marsh	Peter	x102	pmarsh@classicmodelcars.com	6	1,088	Sales Rep
20	1,619	King	Tom	x103	tking@classicmodelcars.com	6	1,088	Sales Rep
21	1,621	Nishi	Mami	x101	mnishi@classicmodelcars.com	5	1,056	Sales Rep
22	1,625	Kato	Yoshimi	x102	ykato@classicmodelcars.com	5	1,621	Sales Rep
23	1,702	Gerard	Martin	x2312	mgerard@classicmodelcars.com	4	1,102	Sales Rep

5. Get all offices

SQL Query:

```sql
SELECT * FROM offices;
```

Query Result:

	officeCode	city	phone	addressLine1	addressLine2	state	country	postalCode	territory
1	1	San Francisco	+1 650 219 4782	100 Market Street	Suite 300	CA	USA	94080	NA
2	2	Boston	+1 215 837 0825	1550 Court Place	Suite 102	MA	USA	02107	NA
3	3	NYC	+1 212 555 3000	523 East 53rd Street	apt. 5A	NY	USA	10022	NA
4	4	Paris	+33 14 723 4404	43 Rue Jouffroy Dabbans	[NULL]	[NULL]	France	75017	EMEA
5	5	Tokyo	+81 33 224 5000	4-1 Kioicho	[NULL]	Chiyoda-Ku	Japan	102-8578	Japan
6	6	Sydney	+61 2 9264 2451	5-11 Wentworth Avenue	Floor #2	[NULL]	Australia	NSW 2010	APAC
7	7	London	+44 20 7877 2041	25 Old Broad Street	Level 7	[NULL]	UK	EC2N 1HN	EMEA

6. Show all product lines

SQL Query:

```sql
SELECT * FROM productlines;
```

Query Result:

	productLine	textDescription	htmlDescription	image
1	Classic Cars	Attention car enthusiasts: Make your wildest car ownership dreams come true. Whether you are looking for classic muscle cars, dream sports cars or movie-inspired miniatures, you	[NULL]	[NULL]
2	Motorcycles	Our motorcycles are state of the art replicas of classic as well as contemporary motorcycle legends such as Harley Davidson, Ducati and Vespa. Models contain stunning details such	[NULL]	[NULL]
3	Planes	Unique, diecast airplane and helicopter replicas suitable for collections, as well as home, office or classroom decorations. Models contain stunning details such as official logos and in	[NULL]	[NULL]
4	Ships	The perfect holiday or anniversary gift for executives, clients, friends, and family. These handcrafted model ships are unique, stunning works of art that will be treasured for generatic	[NULL]	[NULL]
5	Trains	Model trains are a rewarding hobby for enthusiasts of all ages. Whether you are looking for collectible wooden trains, electric streetcars or locomotives, youwill find a number of grea	[NULL]	[NULL]
6	Trucks and Buses	The Truck and Bus models are realistic replicas of buses and specialized trucks produced from the early 1920s to present. The models range in size from 1:12 to 1:50 scale and incluc	[NULL]	[NULL]
7	Vintage Cars	Our Vintage Car models realistically portray automobiles produced from the early 1900s through the 1940s. Materials used include Bakelite, diecast, plastic and wood. Most of the re|	[NULL]	[NULL]

7. List all payments

SQL Query:

```sql
SELECT * FROM payments;
```

Query Result:

	customerNumber	checkNumber	paymentDate	amount
1	103	HQ336336	2004-10-19	6,066.78
2	103	JM555205	2003-06-05	14,571.44
3	103	OM314933	2004-12-18	1,676.14
4	112	BO864823	2004-12-17	14,191.12
5	112	HQ55022	2003-06-06	32,641.98
6	112	ND748579	2004-08-20	33,347.88
7	114	GG31455	2003-05-20	45,864.03
8	114	MA765515	2004-12-15	82,261.22
9	114	NP603840	2003-05-31	7,565.08
10	114	NR27552	2004-03-10	44,894.74
11	119	DB933704	2004-11-14	19,501.82
12	119	LN373447	2004-08-08	47,924.19
13	119	NG94694	2005-02-22	49,523.67
14	121	DB889831	2003-02-16	50,218.95
15	121	FD317790	2003-10-28	1,491.38
16	121	KI831359	2004-11-04	17,876.32
17	121	MA302151	2004-11-28	34,638.14
18	124	AE215433	2005-03-05	101,244.59
19	124	BG255406	2004-08-28	85,410.87
20	124	CQ287967	2003-04-11	11,044.3
21	124	ET64396	2005-04-16	83,598.04
22	124	HI366474	2004-12-27	47,142.7
23	124	HR86578	2004-11-02	55,639.66
24	124	KI131716	2003-08-15	111,654.4
25	124	LF217299	2004-03-26	43,369.3
26	124	NT141748	2003-11-25	45,084.38
27	128	DI925118	2003-01-28	10,549.01
28	128	FA465482	2003-10-18	24,101.81
29	128	FH668230	2004-03-24	33,820.62

8. Get product names and prices

SQL Query:

```sql
SELECT "productName","buyPrice" FROM products;
```

Query Result:

	productName	buyPrice
1	1969 Harley Davidson Ultimate Chopper	48.81
2	1952 Alpine Renault 1300	98.58
3	1996 Moto Guzzi 1100i	68.99
4	2003 Harley-Davidson Eagle Drag Bike	91.02
5	1972 Alfa Romeo GTA	85.68
6	1962 LanciaA Delta 16V	103.42
7	1968 Ford Mustang	95.34
8	2001 Ferrari Enzo	95.59
9	1958 Setra Bus	77.9
10	2002 Suzuki XREO	66.27
11	1969 Corvair Monza	89.14
12	1968 Dodge Charger	75.16
13	1969 Ford Falcon	83.05
14	1970 Plymouth Hemi Cuda	31.92
15	1957 Chevy Pickup	55.7
16	1969 Dodge Charger	58.73
17	1940 Ford Pickup Truck	58.33
18	1993 Mazda RX-7	83.51
19	1937 Lincoln Berline	60.62
20	1936 Mercedes-Benz 500K Special Roads	24.26
21	1965 Aston Martin DB5	65.96
22	1980s Black Hawk Helicopter	77.27
23	1917 Grand Touring Sedan	86.7
24	1948 Porsche 356-A Roadster	53.9
25	1995 Honda Civic	93.89
26	1998 Chrysler Plymouth Prowler	101.51
27	1911 Ford Town Car	33.3
28	1964 Mercedes Tour Bus	74.86
29	1932 Model A Ford J-Coupe	58.48

9. Get customer names and cities

SQL Query:

```sql
SELECT "customerName", city FROM customers;
```

Query Result:

	customerName	city
1	Atelier graphique	Nantes
2	Signal Gift Stores	Las Vegas
3	Australian Collectors, Co.	Melbourne
4	La Rochelle Gifts	Nantes
5	Baane Mini Imports	Stavern
6	Mini Gifts Distributors Ltd.	San Rafael
7	Havel & Zbyszek Co	Warszawa
8	Blauer See Auto, Co.	Frankfurt
9	Mini Wheels Co.	San Francisco
10	Land of Toys Inc.	NYC
11	Euro+ Shopping Channel	Madrid
12	Volvo Model Replicas, Co	Luleå
13	Danish Wholesale Imports	Kobenhavn
14	Saveley & Henriot, Co.	Lyon
15	Dragon Souveniers, Ltd.	Singapore
16	Muscle Machine Inc	NYC
17	Diecast Classics Inc.	Allentown
18	Technics Stores Inc.	Burlingame
19	Handji Gifts& Co	Singapore
20	Herkku Gifts	Bergen
21	American Souvenirs Inc	New Haven
22	Porto Imports Co.	Lisboa
23	Daedalus Designs Imports	Lille
24	La Corne Dabondance, Co.	Paris
25	Cambridge Collectables Co.	Cambridge
26	Gift Depot Inc.	Bridgewater
27	Osaka Souveniers Co.	Kita-ku
28	Vitachrome Inc.	NYC
29	Toys of Finland, Co.	Helsinki

10. List employee first and last names

SQL Query:

```sql
SELECT "firstName", "lastName" FROM employees;
```

Query Result:

	firstName	lastName
1	Diane	Murphy
2	Mary	Patterson
3	Jeff	Firrelli
4	William	Patterson
5	Gerard	Bondur
6	Anthony	Bow
7	Leslie	Jennings
8	Leslie	Thompson
9	Julie	Firrelli
10	Steve	Patterson
11	Foon Yue	Tseng
12	George	Vanauf
13	Loui	Bondur
14	Gerard	Hernandez
15	Pamela	Castillo
16	Larry	Bott
17	Barry	Jones
18	Andy	Fixter
19	Peter	Marsh
20	Tom	King
21	Mami	Nishi
22	Yoshimi	Kato
23	Martin	Gerard

11. Get all order dates

SQL Query:

```sql
SELECT "orderDate" FROM orders;
```

Query Result:

	orderDate
1	2003-01-06
2	2003-01-09
3	2003-01-10
4	2003-01-29
5	2003-01-31
6	2003-02-11
7	2003-02-17
8	2003-02-24
9	2003-03-03
10	2003-03-10
11	2003-03-18
12	2003-03-25
13	2003-03-24
14	2003-03-26
15	2003-04-01
16	2003-04-04
17	2003-04-11
18	2003-04-16
19	2003-04-21
20	2003-04-28
21	2003-04-29
22	2003-05-07
23	2003-05-08
24	2003-05-20
25	2003-05-21
26	2003-05-21

12. Show product vendor list

SQL Query:

```sql
SELECT DISTINCT "productVendor" FROM products;
```

Query Result:

	productVendor
1	Red Start Diecast
2	Highway 66 Mini Classics
3	Min Lin Diecast
4	Studio M Art Models
5	Gearbox Collectibles
6	Unimax Art Galleries
7	Carousel DieCast Legends
8	Exoto Designs
9	Autoart Studio Design
10	Second Gear Diecast
11	Motor City Art Classics
12	Welly Diecast Productions
13	Classic Metal Creations

13. Get all product codes

SQL Query:

```
SELECT "productCode" FROM products;
```

Query Result:

	productCode
1	S10_1678
2	S10_1949
3	S10_2016
4	S10_4698
5	S10_4757
6	S10_4962
7	S12_1099
8	S12_1108
9	S12_1666
10	S12_2823
11	S12_3148
12	S12_3380
13	S12_3891

14. List all countries from offices

SQL Query:

```
SELECT DISTINCT country FROM offices;
```

Query Result:

	country
1	USA
2	France
3	Japan
4	UK
5	Australia

15. Show all order statuses

SQL Query:

```
SELECT DISTINCT status FROM orders;
```

Query Result:

	status
1	Shipped
2	In Process
3	Disputed
4	Resolved
5	On Hold
6	Cancelled

16. Get all payment amounts

SQL Query:

```
SELECT amount FROM payments;
```

Query Result:

	amount
1	6,066.78
2	14,571.44
3	1,676.14
4	14,191.12
5	32,641.98
6	33,347.88
7	45,864.03
8	82,261.22
9	7,565.08
10	44,894.74
11	19,501.82
12	47,924.19
13	49,523.67

17. List all job titles

SQL Query:

```
SELECT DISTINCT "jobTitle" FROM employees;
```

Query Result:

	jobTitle
1	VP Sales
2	Sales Manager (APAC)
3	Sale Manager (EMEA)
4	VP Marketing
5	Sales Rep
6	Sales Manager (NA)
7	President

18. Get customer phone numbers

SQL Query:

```
SELECT phone FROM customers;
```

Query Result:

	phone
1	40.32.2555
2	7025551838
3	03 9520 4555
4	40.67.8555
5	07-98 9555
6	4155551450
7	(26) 642-7555
8	+49 69 66 90 2555
9	6505555787
10	2125557818
11	(91) 555 94 44
12	0921-12 3555
13	31 12 3555

19. Show product MSRP values

SQL Query:

```
SELECT "productName", "MSRP" FROM products;
```

Query Result:

	productName	MSRP
1	1969 Harley Davidson Ultimate Chopper	95.7
2	1952 Alpine Renault 1300	214.3
3	1996 Moto Guzzi 1100i	118.94
4	2003 Harley-Davidson Eagle Drag Bike	193.66
5	1972 Alfa Romeo GTA	136
6	1962 LanciaA Delta 16V	147.74
7	1968 Ford Mustang	194.57
8	2001 Ferrari Enzo	207.8
9	1958 Setra Bus	136.67
10	2002 Suzuki XREO	150.62
11	1969 Corvair Monza	151.08
12	1968 Dodge Charger	117.44
13	1969 Ford Falcon	173.02
14	1970 Plymouth Hemi Cuda	79.8
15	1957 Chevy Pickup	118.5

20. List order numbers

SQL Query:

```
SELECT "orderNumber" FROM orders;
```

Query Result:

	orderNumber
1	10,100
2	10,101
3	10,102
4	10,103
5	10,104
6	10,105
7	10,106
8	10,107
9	10,108
10	10,109
11	10,110
12	10,111
13	10,112
14	10,113
15	10,114

21. Get orders with customer names

SQL Query:

```
SELECT o."orderNumber", c."customerName"
FROM orders o
JOIN customers c ON o."customerNumber" = c."customerNumber";
```

Query Result:

	orderNumber	customerName
1	10,100	Online Diecast Creations Co.
2	10,101	Blauer See Auto, Co.
3	10,102	Vitachrome Inc.
4	10,103	Baane Mini Imports
5	10,104	Euro+ Shopping Channel
6	10,105	Danish Wholesale Imports
7	10,106	Rovelli Gifts
8	10,107	Land of Toys Inc.
9	10,108	Cruz & Sons Co.
10	10,109	Motor Mint Distributors Inc.
11	10,110	AV Stores, Co.
12	10,111	Mini Wheels Co.
13	10,112	Volvo Model Replicas, Co
14	10,113	Mini Gifts Distributors Ltd.
15	10,114	La Corne Dabondance, Co.

22. Get employees with office city

SQL Query:

```
SELECT e."firstName", e."lastName", o.city
FROM employees e
```

```sql
JOIN offices o ON e."officeCode" = o."officeCode";
```

Query Result:

	firstName	lastName	city
1	Diane	Murphy	San Francisco
2	Mary	Patterson	San Francisco
3	Jeff	Firrelli	San Francisco
4	William	Patterson	Sydney
5	Gerard	Bondur	Paris
6	Anthony	Bow	San Francisco
7	Leslie	Jennings	San Francisco
8	Leslie	Thompson	San Francisco
9	Julie	Firrelli	Boston
10	Steve	Patterson	Boston
11	Foon Yue	Tseng	NYC
12	George	Vanauf	NYC
13	Loui	Bondur	Paris
14	Gerard	Hernandez	Paris
15	Pamela	Castillo	Paris

23. Get payments with customer names

SQL Query:

```sql
SELECT p.amount, c."customerName"
FROM payments p
JOIN customers c ON p."customerNumber" = c."customerNumber";
```

Query Result:

	amount	customerName
1	6,066.78	Atelier graphique
2	14,571.44	Atelier graphique
3	1,676.14	Atelier graphique
4	14,191.12	Signal Gift Stores
5	32,641.98	Signal Gift Stores
6	33,347.88	Signal Gift Stores
7	45,864.03	Australian Collectors, Co.
8	82,261.22	Australian Collectors, Co.
9	7,565.08	Australian Collectors, Co.
10	44,894.74	Australian Collectors, Co.
11	19,501.82	La Rochelle Gifts
12	47,924.19	La Rochelle Gifts
13	49,523.67	La Rochelle Gifts
14	50,218.95	Baane Mini Imports
15	1,491.38	Baane Mini Imports

24. Get order details with product names

SQL Query:

```sql
SELECT od."orderNumber", p."productName"
FROM orderdetails od
JOIN products p ON od."productCode" = p."productCode";
```

Query Result:

	orderNumber	productName
1	10,100	1917 Grand Touring Sedan
2	10,100	1911 Ford Town Car
3	10,100	1932 Alfa Romeo 8C2300 Spider Sport
4	10,100	1936 Mercedes Benz 500k Roadster
5	10,101	1932 Model A Ford J-Coupe
6	10,101	1928 Mercedes-Benz SSK
7	10,101	1939 Chevrolet Deluxe Coupe
8	10,101	1938 Cadillac V-16 Presidential Limousine
9	10,102	1937 Lincoln Berline
10	10,102	1936 Mercedes-Benz 500K Special Roadster
11	10,103	1952 Alpine Renault 1300
12	10,103	1962 LanciaA Delta 16V
13	10,103	1958 Setra Bus
14	10,103	1940 Ford Pickup Truck
15	10,103	1926 Ford Fire Engine

25. Get products with product line description

SQL Query:

```sql
SELECT p."productName", pl."textDescription"
FROM products p
JOIN productlines pl ON p."productLine" = pl."productLine";
```

Query Result:

	productName	textDescription
1	1969 Harley Davidson Ultimate Chopper	Our motorcycles are state of the art replicas of classic as well as contemporary motorcycle legends such as Harley Davidson, Ducati and Vespa. Models contain stunning details such as official logos, rotating wheels, working kickstand
2	1952 Alpine Renault 1300	Attention car enthusiasts: Make your wildest car ownership dreams come true. Whether you are looking for classic muscle cars, dream sports cars or movie-inspired miniatures, you will find great choices in this category. These replic
3	1996 Moto Guzzi 1100i	Our motorcycles are state of the art replicas of classic as well as contemporary motorcycle legends such as Harley Davidson, Ducati and Vespa. Models contain stunning details such as official logos, rotating wheels, working kickstand
4	2003 Harley-Davidson Eagle Drag Bike	Our motorcycles are state of the art replicas of classic as well as contemporary motorcycle legends such as Harley Davidson, Ducati and Vespa. Models contain stunning details such as official logos, rotating wheels, working kickstand
5	1972 Alfa Romeo GTA	Attention car enthusiasts: Make your wildest car ownership dreams come true. Whether you are looking for classic muscle cars, dream sports cars or movie-inspired miniatures, you will find great choices in this category. These replic
6	1962 LanciaA Delta 16V	Attention car enthusiasts: Make your wildest car ownership dreams come true. Whether you are looking for classic muscle cars, dream sports cars or movie-inspired miniatures, you will find great choices in this category. These replic
7	1968 Ford Mustang	Attention car enthusiasts: Make your wildest car ownership dreams come true. Whether you are looking for classic muscle cars, dream sports cars or movie-inspired miniatures, you will find great choices in this category. These replic
8	2001 Ferrari Enzo	Attention car enthusiasts: Make your wildest car ownership dreams come true. Whether you are looking for classic muscle cars, dream sports cars or movie-inspired miniatures, you will find great choices in this category. These replic
9	1958 Setra Bus	The Truck and Bus models are realistic replicas of buses and specialized trucks produced from the early 1920s to present. The models range in size from 1:12 to 1:50 scale and include numerous limited edition and several out-of-prod
10	2002 Suzuki XREO	Our motorcycles are state of the art replicas of classic as well as contemporary motorcycle legends such as Harley Davidson, Ducati and Vespa. Models contain stunning details such as official logos, rotating wheels, working kickstand
11	1969 Corvair Monza	Attention car enthusiasts: Make your wildest car ownership dreams come true. Whether you are looking for classic muscle cars, dream sports cars or movie-inspired miniatures, you will find great choices in this category. These replic
12	1968 Dodge Charger	Attention car enthusiasts: Make your wildest car ownership dreams come true. Whether you are looking for classic muscle cars, dream sports cars or movie-inspired miniatures, you will find great choices in this category. These replic
13	1969 Ford Falcon	Attention car enthusiasts: Make your wildest car ownership dreams come true. Whether you are looking for classic muscle cars, dream sports cars or movie-inspired miniatures, you will find great choices in this category. These replic
14	1970 Plymouth Hemi Cuda	Attention car enthusiasts: Make your wildest car ownership dreams come true. Whether you are looking for classic muscle cars, dream sports cars or movie-inspired miniatures, you will find great choices in this category. These replic

26. Get customers with sales rep names

SQL Query:

```sql
SELECT c."customerName", e."firstName", e."lastName"
FROM customers c
JOIN employees e ON c."salesRepEmployeeNumber" = e."employeeNumber";
```

Query Result:

	customerName	firstName	lastName
1	Atelier graphique	Gerard	Hernandez
2	Signal Gift Stores	Leslie	Thompson
3	Australian Collectors, Co.	Andy	Fixter
4	La Rochelle Gifts	Gerard	Hernandez
5	Baane Mini Imports	Barry	Jones
6	Mini Gifts Distributors Ltd.	Leslie	Jennings
7	Blauer See Auto, Co.	Barry	Jones
8	Mini Wheels Co.	Leslie	Jennings
9	Land of Toys Inc.	George	Vanauf
10	Euro+ Shopping Channel	Gerard	Hernandez
11	Volvo Model Replicas, Co	Barry	Jones
12	Danish Wholesale Imports	Pamela	Castillo

27. Get orders with customer city

SQL Query:

```sql
SELECT o."orderNumber", c.city
FROM orders o
JOIN customers c ON o."customerNumber" = c."customerNumber";
```

Query Result:

	orderNumber	city
1	10,100	Nashua
2	10,101	Frankfurt
3	10,102	NYC
4	10,103	Stavern
5	10,104	Madrid
6	10,105	Kobenhavn
7	10,106	Bergamo
8	10,107	NYC
9	10,108	Makati City
10	10,109	Philadelphia
11	10,110	Manchester
12	10,111	San Francisco

28. Get employees and their manager

SQL Query:

```sql
SELECT e."firstName", m."firstName" AS manager
FROM employees e
LEFT JOIN employees m ON e."reportsTo" = m."employeeNumber";
```

Query Result:

	firstName	manager
1	Diane	[NULL]
2	Mary	Diane
3	Jeff	Diane
4	William	Mary
5	Gerard	Mary
6	Anthony	Mary
7	Leslie	Anthony
8	Leslie	Anthony
9	Julie	Anthony
10	Steve	Anthony
11	Foon Yue	Anthony
12	George	Anthony

29. Get orderdetails with product vendor

SQL Query:

```
SELECT od."orderNumber", p."productVendor"
FROM orderdetails od
JOIN products p ON od."productCode" = p."productCode";
```

Query Result:

	orderNumber	productVendor
1	10,100	Welly Diecast Productions
2	10,100	Motor City Art Classics
3	10,100	Exoto Designs
4	10,100	Red Start Diecast
5	10,101	Autoart Studio Design
6	10,101	Gearbox Collectibles
7	10,101	Motor City Art Classics
8	10,101	Classic Metal Creations
9	10,102	Motor City Art Classics
10	10,102	Studio M Art Models
11	10,103	Classic Metal Creations
12	10,103	Second Gear Diecast

30. Get payments with customer country

SQL Query:

```
SELECT p.amount, c.country
FROM payments p
JOIN customers c ON p."customerNumber" = c."customerNumber";
```

Query Result:

	amount	country
1	6,066.78	France
2	14,571.44	France
3	1,676.14	France
4	14,191.12	USA
5	32,641.98	USA
6	33,347.88	USA
7	45,864.03	Australia
8	82,261.22	Australia
9	7,565.08	Australia
10	44,894.74	Australia
11	19,501.82	France
12	47,924.19	France

31. Count customers per country

SQL Query:

```
SELECT country, COUNT(*) FROM customers GROUP BY country;
```

Query Result:

	country	count
1	Italy	4
2	Sweden	2
3	USA	36
4	Ireland	2
5	Germany	13
6	Singapore	3
7	Canada	3
8	Finland	3
9	Portugal	2
10	France	12
11	Israel	1
12	South Africa	1

32. Total payments per customer

SQL Query:

```
SELECT "customerNumber", SUM(amount)
FROM payments
GROUP BY "customerNumber";
```

Query Result:

	customerNumber	sum
1	455	70,378.65
2	448	76,776.44
3	146	130,305.35
4	350	71,547.53
5	382	85,060
6	314	62,253.85
7	386	90,143.31
8	278	127,529.69
9	450	59,551.38
10	424	69,214.33
11	406	86,436.97
12	462	88,627.49

33. Number of orders per status

SQL Query:

```
SELECT status, COUNT(*) FROM orders GROUP BY status;
```

Query Result:

	status	count
1	Shipped	303
2	In Process	6
3	Disputed	3
4	Resolved	4
5	On Hold	4
6	Cancelled	6

34. Products per product line

SQL Query:

```
SELECT "productLine", COUNT(*) FROM products GROUP BY "productLine";
```

Query Result:

	productLine	count
1	Motorcycles	13
2	Ships	9
3	Planes	12
4	Classic Cars	38
5	Trains	3
6	Trucks and Buses	11
7	Vintage Cars	24

35. Employees per office

SQL Query:

```
SELECT "officeCode", COUNT(*) FROM employees GROUP BY "officeCode";
```

Query Result:

	officeCode	count
1	6	4
2	2	2
3	4	5
4	7	2
5	5	2
6	1	6
7	3	2

36. Total stock per product vendor

SQL Query:

```
SELECT "productVendor", SUM("quantityInStock")
FROM products
GROUP BY "productVendor";
```

Query Result:

	productVendor	sum
1	Red Start Diecast	35,046
2	Highway 66 Mini Classics	37,520
3	Min Lin Diecast	50,089
4	Studio M Art Models	42,253
5	Gearbox Collectibles	60,495
6	Unimax Art Galleries	38,191
7	Carousel DieCast Legends	40,805
8	Exoto Designs	44,166
9	Autoart Studio Design	30,093
10	Second Gear Diecast	42,865
11	Motor City Art Classics	43,105
12	Welly Diecast Productions	45,095
13	Classic Metal Creations	45,408

37. Average buy price per product line

SQL Query:

```
SELECT "productLine", AVG("buyPrice")
FROM products
GROUP BY "productLine";
```

Query Result:

	productLine	avg
1	Motorcycles	50.6853846154
2	Ships	47.0077777778
3	Planes	49.6291666667
4	Classic Cars	64.4463157895
5	Trains	43.9233333333
6	Trucks and Buses	56.3290909091
7	Vintage Cars	46.06625

38. Orders per customer

SQL Query:

```
SELECT "customerNumber", COUNT(*)
FROM ordersD
GROUP BY "customerNumber";
```

Query Result:

	customerNumber	count
1	455	2
2	448	3
3	146	3
4	350	3
5	382	4
6	314	3
7	386	3
8	278	3
9	450	4
10	406	3
11	424	3

39. Max MSRP per product line

SQL Query:

```
SELECT "productLine", MAX("MSRP")
FROM products
GROUP BY "productLine";
```

Query Result:

	productLine	max
1	Motorcycles	193.66
2	Ships	122.89
3	Planes	157.69
4	Classic Cars	214.3
5	Trains	100.84
6	Trucks and Buses	136.67
7	Vintage Cars	170

40. Min buy price per vendor

SQL Query:

```
SELECT "productVendor", MIN("buyPrice")
FROM products
GROUP BY "productVendor";
```

Query Result:

	productVendor	min
1	Red Start Diecast	21.75
2	Highway 66 Mini Classics	32.37
3	Min Lin Diecast	34.35
4	Studio M Art Models	23.14
5	Gearbox Collectibles	24.14
6	Unimax Art Galleries	33.3
7	Carousel DieCast Legends	15.91
8	Exoto Designs	43.26
9	Autoart Studio Design	26.3
10	Second Gear Diecast	16.24
11	Motor City Art Classics	22.57

41. Total number of customers

SQL Query:

```
SELECT COUNT(*) FROM customers;
```

Query Result:

	count
1	122

42. Total number of products

SQL Query:

```
SELECT COUNT(*) FROM products;
```

Query Result:

	count
1	110

43. Total revenue from payments

SQL Query:

```
SELECT SUM(amount) FROM payments;
```

Query Result:

	sum
1	8,853,839.23

44. Average product price

SQL Query:

```
SELECT AVG("buyPrice") FROM products;
```

Query Result:

	avg
1	54.3951818182

45. Max payment amount

SQL Query:

```
SELECT MAX(amount) FROM payments;
```

Query Result:

	max
1	120,166.58

46. Min payment amount

SQL Query:

```
SELECT MIN(amount) FROM payments;
```

Query Result:

	min
1	615.45

47. Count total orders

SQL Query:

```
SELECT COUNT(*) FROM orders;
```

Query Result:

	count
1	326

48. Total quantity in stock

SQL Query:

```
SELECT SUM("quantityInStock") FROM products;
```

Query Result:

	sum
1	555,131

49. Average MSRP

SQL Query:

```sql
SELECT AVG("MSRP") FROM products;
```

Query Result:

	avg
1	100.4387272727

50. Number of employees

SQL Query:

```sql
SELECT COUNT(*) FROM employees;
```

Query Result:

	count
1	23

EXPLANATIONS

1. This query uses the `SELECT *` command, which acts as a wildcard to retrieve every single column available in the table. We are specifically querying the `products` table to see the complete inventory catalog. It is the most basic form of data retrieval, perfect for getting a bird's-eye view of all the product data stored in your database before filtering it down.

2. Similar to the previous query, the asterisk (`*`) tells the database to fetch all columns without filtering anything out. By pointing it `FROM customers`, we are asking to see the entire list of people or companies who have registered in our system. This will return a massive table showing everything from customer names to their contact details and credit limits.

3. This statement grabs all the data housed inside the `orders` table. Because we use the `*` wildcard, the result will include every piece of information about every transaction. You will see columns for order dates, statuses, and customer reference numbers, giving a complete chronological history of all purchasing activity.

4. Here, we are pulling the full roster of staff members by querying the `employees` table. The `SELECT *` ensures that we don't miss any columns, meaning the output will show employee IDs, names, job titles, and exactly who they report to. It is the quickest way to pull up the entire organizational directory in one single execution.

5. This command fetches every single row and column from the `offices` table. It provides a complete overview of all the physical locations or branches associated with the company. The output will detail the city, phone numbers, state, and exact postal addresses for every office worldwide.

6. By querying the `productlines` table with a `SELECT *`, we are retrieving all the broad categories of products the company sells. This table usually contains the category name (like 'Classic Cars' or 'Motorcycles') along with a detailed text or HTML description of that line. It gives a high-level view of the company's major business sectors.

7. This query pulls up the entire financial ledger directly from the `payments` table. Using the wildcard `*` means we will see the customer number, the check reference number, the date the payment cleared, and the exact monetary amount. It is essentially generating a raw, unfiltered report of all incoming revenue.

8. Instead of grabbing everything, this query specifically names the columns we want: `productName` and `buyPrice`. By doing this, we filter out all the extra noise (like product descriptions or vendor names) from the `products` table. The result is a clean, easy-to-read list showing exactly what each item is called and how much it costs the company to acquire.

9. This statement narrows down the data from the `customers` table to just two specific pieces of information. It selects the `customerName` and the `city` they are located in, deliberately ignoring things like phone numbers and credit limits. This is highly useful for quickly checking where your primary client base is geographically located.

10. Here we are formatting a simple employee directory by explicitly requesting only the `firstName` and `lastName` columns. By reading from the `employees` table, it strips away their job titles, office codes, and internal email addresses. The final output is a straightforward, human-readable list of the names of everyone working at the company.

11. This query isolates a single column, `orderDate`, from the broader `orders` table. It will return a vertical list of every date on which an order was placed, without showing who placed it or what they actually bought. This type of query is often the very first step when a data analyst wants to look at order frequency or seasonality over time.

12. This query introduces the `DISTINCT` keyword, which is incredibly powerful for removing duplicate entries. Instead of listing the vendor for every single product (which would repeat names constantly), it scans the `products` table and returns a clean list of unique vendor names. It essentially answers the question, "Which unique companies supply our inventory?"

13. By selecting only the `productCode` from the `products` table, this query generates a list of unique alphanumeric identifiers for every item in the catalog. This is particularly useful for developers or database administrators who need just the primary keys to link to other tables. It strips away all descriptive data, leaving only the internal ID numbers.

14. Just like the vendor query, this uses `DISTINCT` to filter out duplicate geographic values, this time from the `offices` table. If the company has five offices in the USA and three in France, this query will only print "USA" and "France" exactly once. It provides a quick, consolidated summary of the international footprint of the company.

15. This query is perfect for understanding the different operational stages an order can be in. By using `DISTINCT` on the `status` column of the `orders` table, it prevents the database from printing "Shipped" thousands of times. Instead, you get a short, unique list of all possible workflow statuses, such as Shipped, Pending, Cancelled, or On Hold.

16. This simple query isolates the numerical monetary values from the `payments` table. It retrieves only the `amount` column, completely ignoring who made the payment or when the transaction happened. While a bit raw on its own, fetching a pure list of numbers like this is often used before exporting data to Excel or Python for statistical analysis.

17. Using the `DISTINCT` keyword again, this query scans the `employees` table to find all the unique roles within the company. Instead of seeing "Sales Rep" listed a hundred times for each individual employee, it collapses them into a single entry. This gives human resources or management a quick overview of the organizational structure and available positions.

18. This statement retrieves just the `phone` column from the `customers` table. It creates a vertical list of every customer's contact number, leaving out their names, credit limits, or addresses. A query like this would be exactly what a marketing or sales team might use to generate a raw call list for a new outreach campaign.

19. Here we are pulling two specific columns from the `products` table: the item's name (`productName`) and its Manufacturer's Suggested Retail Price (`MSRP`). This creates a clean, two-column price list for the entire inventory without any internal cost data. It is an excellent query for a sales team that needs to know the exact retail price tag they should quote to clients.

20. This command fetches only the `orderNumber` column from the `orders` table. Because `orderNumber` is the primary key for that table, this query effectively generates a complete list of every transaction ID ever recorded. It strips away the dates, statuses, and customer details, providing just the raw list of system reference numbers.

21. This is where relational databases shine! It uses an `INNER JOIN` to stitch together the `orders` (aliased as 'o') and `customers` (aliased as 'c') tables. It lines up the rows where the `customerNumber` matches perfectly in both tables. The result is a neat list showing the specific order number right next to the actual name of the person or company who placed it.

22. This query links the `employees` table to the `offices` table to find out exactly where everyone is stationed. By joining them on the shared `officeCode` column, the database can definitively match the employee to their physical building. The output presents the employee's first and last name right next to the city of their assigned office branch.

23. To figure out who is actually paying the bills, this query joins the `payments` table with the `customers` table. It uses the `customerNumber` as the critical bridge between the two datasets. Instead of just seeing a payment amount next to a meaningless ID number, you now see the financial amount paired directly with the client's actual name.

24. This query bridges the gap between a specific order's line items and the actual product catalog. By joining the `orderdetails` table and the `products` table on the `productCode`, it translates the internal item codes into readable data. You get an output showing the order number directly alongside the human-readable name of the product that was purchased.

25. Here, we are enriching our product data by pulling in descriptive information from the `productlines` table. The query joins the two tables using the shared `productLine` category name. This allows the output to display the specific `productName` alongside the longer, detailed `textDescription` of the broader category it belongs to.

26. This query maps out the specific relationship between clients and their internal account managers. It joins the `customers` table to the `employees` table by matching the customer's assigned `salesRepEmployeeNumber` to the employee's actual `employeeNumber`. The result is a clean list pairing every customer with the exact first and last name of the sales rep looking after them.

27. If you want to know where your orders are originating geographically, this is the perfect query. It joins the `orders` table with the `customers` table using the `customerNumber` as the link. It then displays the unique `orderNumber` alongside the `city` that the purchasing customer is based in, which is highly useful for shipping and logistical analysis.

28. This is a special type of query called a "Self Join," where the `employees` table is joined to itself to map out the corporate hierarchy. It matches the `reportsTo` ID of an employee with the `employeeNumber` of their boss. The `LEFT JOIN` ensures that even the top executive (who reports to no one) still shows up on the list, displaying the employee's name next to their manager's name.

29. This query helps trace a specific order back to the original external suppliers. It connects the line items in the `orderdetails` table to the main `products` table using the `productCode`. The final result shows the `orderNumber` paired with the `productVendor`, telling you exactly which third-party companies supplied the physical goods for that specific transaction.

30. This query is perfect for analyzing international revenue streams and global cash flow. It joins the `payments` table to the `customers` table based on the shared `customerNumber`. By selecting the payment `amount` and the customer's `country`, you can easily see a breakdown of individual transactions paired with the geographical region the money came from.

31. This introduces the `GROUP BY` clause, which is essential for data aggregation. It gathers all the rows in the `customers` table and sorts them into distinct groups based on their `country`. The `COUNT(*)` function then counts how many individual customer records fall into each specific country bucket, giving you a summary of your customer base by region.

32. Here, we are calculating the lifetime financial value of our clients. The query groups the `payments` table by the unique `customerNumber`. Instead of just counting rows, it uses the `SUM(amount)` mathematical function to add up all the individual payments made by that specific customer, returning their total historical spend with the company.

33. This query is a fantastic way to check the overall health and backlog of your fulfillment pipeline. It groups every row in the `orders` table by its current `status` (like 'Shipped', 'In Process', or 'Cancelled'). The `COUNT(*)` function then tallies up exactly how many orders are currently sitting in each of those operational stages.

34. To understand how your inventory is distributed across categories, this query organizes the `products` table into groups based on the `productLine`. The `COUNT(*)` function then calculates how many individual product models belong to each specific category. It quickly shows whether your catalog is dominated by 'Classic Cars' or 'Motorcycles'.

35. This query acts as a quick, aggregated headcount tool for the company's various physical branches. It takes the `employees` table and groups everyone by the `officeCode` they are assigned to. By using `COUNT(*)`, it calculates the exact number of staff members currently working out of each office location.

36. This is a great logistics query that groups the `products` table based on the `productVendor`. By applying the `SUM()` function to the `quantityInStock` column, it aggregates the data. It adds up the total number of physical items currently sitting in the warehouse that were supplied by each specific vendor.

37. This query is highly useful for financial analysis of your inventory procurement costs. It groups the `products` by their broader `productLine` category. Then, it uses the `AVG()` function to calculate the mathematical mean of the `buyPrice` across all items in that category, letting you know which product lines are generally the most expensive to acquire.

38. To find out who your most frequent and loyal shoppers are, this query groups the `orders` table by the `customerNumber`. The `COUNT(*)` function simply tallies up how many individual order records exist for each unique customer ID. It provides a straightforward list showing exactly how many times each client has checked out.

39. This query helps identify the premium, top-tier items in your catalog. It groups the `products` table by the `productLine` category. The `MAX()` function then scans through all the Manufacturer's Suggested Retail Prices (`MSRP`) within each distinct group and extracts only the absolute highest price tag found in that category.

40. If you are looking for the cheapest items supplied by your business partners, this query does the trick. It groups the `products` table by the `productVendor`. The `MIN()` function then evaluates the `buyPrice` column within each vendor's group and returns the absolute lowest cost item they sell to the company.

41. This is one of the simplest but most important top-level aggregation queries. By using `COUNT(*)` without any `GROUP BY` clauses, it simply counts every single row that exists inside the `customers` table. The output is a single, definitive number representing the grand total of registered clients in your entire database.

42. Similar to the customer count, this query calculates the total breadth of your catalog offerings. The `COUNT(*)` function scans the entire `products` table from top to bottom and returns the total number of rows. This gives you a single numerical value representing every unique item the company currently offers for sale.

43. This query calculates the most critical financial metric for any business: total gross revenue. It uses the `SUM()` function to add together every single numerical value present in the `amount` column of the `payments` table. The result is a single massive number representing all the money the company has collected historically.

44. This query calculates the central tendency of your inventory acquisition costs. It applies the `AVG()` mathematical function to the `buyPrice` column across the entire `products` table. The output is a single floating-point number that represents the overall average cost the company pays to acquire an item.

45. To find the single largest financial transaction ever recorded by the system, this query uses the `MAX()` function on the `payments` table. It scans every value in the `amount` column and returns only the absolute highest number. It is a quick way to identify the most lucrative single payment ever received from a client.

46. Conversely, this query looks for the absolute smallest financial transaction in the company's history. The `MIN()` function evaluates all the data in the `amount` column of the `payments` table. It returns the single lowest monetary value, which might represent a tiny account adjustment, a test transaction, or a very small balance payment.

47. This query provides a high-level metric on overall business and sales activity. The `COUNT(*)` function simply tallies up every single row currently residing in the `orders` table. The output is a single figure that tells you the total historical volume of orders placed since the database was first brought online.

48. This is the ultimate inventory health check for warehouse management. It uses the `SUM()` function to add up all the individual numbers listed in the `quantityInStock` column across the entire `products` table. The result is a single large number representing the total physical count of every single item sitting on the warehouse shelves.

49. This query helps establish a baseline for the company's global retail pricing strategy. The `AVG()` function calculates the mathematical mean of the Manufacturer's Suggested Retail Price (`MSRP`) column across all rows in the `products` table. It gives a single number representing the overall average retail price tag of your entire inventory.

50. Finally, this simple aggregation query calculates the total internal headcount of the organization. By applying `COUNT(*)` to the `employees` table, it tallies up every single row without filtering anything out. The output is a single number that tells management exactly how many staff members are currently recorded in the corporate directory.